

Histology, Part One. Patterns of Injury.

Sooner or later, every fellow in paediatric hepatology sits down with a biopsy report and does not know what to do with it. The report is long. It describes portal tracts, lobules, bile, ducts and cells with unfamiliar names. The final line may say something careful, such as "features compatible with" or "cannot exclude". And the clinical team is waiting for you to say what it means.

This chapter is about how to read that report. The key idea is simple. A liver biopsy rarely gives you a diagnosis. It gives you a pattern. A pattern narrows the list of possible causes, and it tells you which test will settle the question. If you know the main patterns, and you know what each one points to, the report stops being a wall of words and becomes a set of clues.

We will look at four groups of patterns. First, where in the liver the injury sits, which we call zonation. Second, interface hepatitis, the pattern behind autoimmune hepatitis and several of its mimics. Third, cholestasis, and in particular where the bile is sitting, which is the most useful clue in a jaundiced infant. Fourth, the loss of bile ducts, which we call ductopenia. At the end we will look at what the major guidelines say, and at four clinical cases.

A note on the sources. For infant cholestasis, the main guideline is the joint North American and European paediatric society guideline of 2017, written by Fawaz and colleagues. For autoimmune liver disease in children, it is the European paediatric society position paper of 2018. For autoimmune hepatitis in general, we will use the American Association for the Study of Liver Diseases guidance of 2019, the European Association for the Study of the Liver guideline of 2025, and the consensus of the International Autoimmune Hepatitis Pathology Group, published in 2022. For the technical side of liver biopsy, we will use the British Society of Gastroenterology guideline of 2020.

Part one. The acinus, not the lobule.

Most of us first learned the liver as a set of hexagons. Each hexagon is a classic lobule, with a central vein in the middle and portal tracts at the corners. This picture is correct, but it describes shape. It does not describe how blood moves, and blood flow is what explains injury.

A more useful picture is the acinus, described by Rappaport. In the acinus, the starting point is the portal tract. Blood enters the liver there, from the portal vein and the hepatic artery. It then flows along the sinusoids, past the plates of liver cells, towards the central vein, where it leaves. The liver cells along this path are

divided into three zones. Zone one is next to the portal tract. Zone three is next to the central vein. Zone two lies in between.

The cells in zone one receive blood first. The oxygen pressure there is about sixty to sixty-five millimetres of mercury. The cells in zone three receive blood last, after every other cell has taken its share, and the oxygen pressure there is only about thirty to thirty-five millimetres of mercury.

The zones do different work. Zone one cells make glucose, burn fat, make urea and take up bile acids from the blood. Zone three cells break down glucose, make fat and carry the main drug-metabolising enzymes, especially the enzyme called CYP two E one. The last one to three layers of cells around the central vein are the only ones that make glutamine synthetase. Pathologists use a stain for this enzyme to find zone three on a slide.

These differences are not fixed by position alone. They are maintained by the oxygen gradient and by signalling pathways, especially Wnt and beta-catenin, and hedgehog. In 2017, Halpern and colleagues used single-cell methods to map gene activity across the acinus. They found that about half of all liver genes are zonated. Some genes are most active not at either end but in the middle, in zone two.

Why does this matter clinically? Because the zone of injury tells you the mechanism before it tells you the cause. If a pathologist says the injury is in zone three, you already know you are looking for something that attacks the cells with the least oxygen or the most drug metabolism. If the injury is in zone one, you are looking for something that enters from the portal tract.

Part two. Where the damage sits.

Let us go through the zones one at a time.

Zone three injury has three classic causes. The first is paracetamol. Paracetamol is converted by CYP two E one into a toxic product called NAPQI. Normally NAPQI is neutralised by glutathione. In zone three, CYP two E one is most active and glutathione is lowest, so this is where the cells die. The second cause is lack of blood flow or poor drainage. Shock, low cardiac output and congestion all hit zone three first, because these cells are last in line for oxygen. In paediatric practice, the child after a Fontan operation is the classic example of congestion. Sinusoidal obstruction syndrome, also called veno-occlusive disease, also injures zone three. The third cause, which is easy to forget, is acute autoimmune hepatitis. In the pooled data tabulated by the American guidance, necrosis around the central vein was present in about twenty-nine percent of patients with

autoimmune hepatitis. So zone three necrosis in a teenager is not always paracetamol.

Zone one injury also has three main patterns. Interface hepatitis starts here, at the limiting plate, the row of liver cells that borders the portal tract. We will come back to it in detail. Chronic cholestasis leaves its marks in zone one too: swollen, pale liver cells, which pathologists call cholate stasis; copper and copper-binding protein; and liver cells that begin to express the bile duct marker CK seven. Iron also loads zone one cells first in genetic iron overload.

Zone two injury is rare. Isolated necrosis in the middle of the acinus is seen in dengue and yellow fever. When you see this pattern, ask about travel.

There is one important paediatric twist. In adults, fatty liver disease with inflammation, which we now call steatohepatitis, usually starts in zone three. In 2005, Schwimmer and colleagues described the biopsies of one hundred children with fatty liver disease. Fifty-one percent had a different pattern, which they called type two. In type two, the fat and inflammation sit in zone one and around the portal tract. Only seventeen percent had the adult-type zone three pattern. Advanced fibrosis was more common with the type two pattern. So a child's steatohepatitis should be read with paediatric eyes. A report that says "not typical of steatohepatitis" because the changes are portal may simply be describing the usual paediatric picture.

Part three. Interface hepatitis.

Interface hepatitis is one of the most important words on a biopsy report. It means that inflammatory cells, mainly lymphocytes and plasma cells, have crossed the limiting plate. They have left the portal tract and are attacking the liver cells at its edge. The older name was piecemeal necrosis.

Interface hepatitis is a pattern, not a diagnosis. In children, five main causes produce it, and the biopsy alone rarely tells them apart.

The first cause is autoimmune hepatitis. Here the portal infiltrate is often rich in plasma cells, there is inflammation inside the lobule as well, and there may be small pink droplets inside Kupffer cells, called hyaline globules. What settles the diagnosis is the rest of the picture: a raised IgG, the autoantibodies, which are antinuclear and smooth muscle antibodies in type one disease and anti liver kidney microsomal type one and anti liver cytosol type one antibodies in type two disease, and finally a response to steroids.

The second cause is autoimmune sclerosing cholangitis. This is the childhood overlap between autoimmune hepatitis and sclerosing cholangitis. The biopsy shows the same interface hepatitis, often with added signs of bile duct injury, a

ductular reaction and copper. What settles it is imaging of the bile ducts, usually a magnetic resonance cholangiogram.

The third cause is Wilson disease. The biopsy may show fat, liver cell nuclei that look empty because they are filled with glycogen, and Mallory-Denk bodies, which are clumps of damaged protein inside liver cells. Copper stains on the slide are unreliable, because copper is patchy and early copper is spread through the cytoplasm where the stain cannot see it. What settles Wilson disease is a measurement of copper in the liver tissue, together with the ceruloplasmin, urine copper and testing of the ATP seven B gene.

The fourth cause is chronic viral hepatitis. In hepatitis B, some liver cells have a smooth, pale, glassy cytoplasm, called ground-glass cells. In hepatitis C, there are lymphoid aggregates in the portal tracts and some fat. Serology and viral load settle the question.

The fifth cause is drug-induced liver injury. Minocycline and nitrofurantoin are the classic drugs. They can produce a picture that is indistinguishable from autoimmune hepatitis. Eosinophils in the infiltrate sometimes point to a drug, but not reliably. What settles it is the drug history and the course after the drug is stopped.

The lesson of this part is that interface hepatitis opens a differential. It does not close one.

Part four. Autoimmune hepatitis on the biopsy.

How often are the classic features of autoimmune hepatitis actually present? The American guidance of 2019 pooled the published literature. Plasma cell infiltration was present in about sixty-six percent of patients. Emperipolesis, where a lymphocyte appears to sit inside a liver cell, was present in about sixty-five percent. Lobular hepatitis was present in forty-seven percent. Rosettes, which are small rings of regenerating liver cells, were present in thirty-three percent. Necrosis around the central vein was present in twenty-nine percent. And in children, cirrhosis was already present at diagnosis in about thirty-eight percent.

Two lessons follow. First, about one biopsy in three from a patient with autoimmune hepatitis has no excess of plasma cells. Plasma cells support the diagnosis, but their absence does not exclude it. Second, more than a third of children are already cirrhotic when the diagnosis is made. Autoimmune hepatitis in childhood is often found late.

For many years, emperipolesis and rosettes were considered typical of autoimmune hepatitis. The simplified diagnostic criteria of 2008, from the

International Autoimmune Hepatitis Group, gave two points for typical histology, and typical histology required them. Two studies changed this view.

In 2017, Balitzer and colleagues at the University of California, San Francisco, reviewed eighty-eight biopsies from patients with autoimmune hepatitis. Under the strict 2008 definition, only eight percent scored as typical. When they used a broader definition, which counted inflammatory activity, plasma cells and even some bile duct changes, seventy-seven percent scored as typical. With this change, about seventeen percent of patients moved into the probable or definite category who had been missed by the original criteria.

In 2018, Gurung and colleagues compared autoimmune hepatitis with hepatitis C, carefully matched for the grade of inflammation and the stage of fibrosis. Once severity was matched, rosettes were no more common in autoimmune hepatitis than in hepatitis C. The P value was one. Emperipolesis was also no different, with a P value of zero point four. In other words, rosettes and emperipolesis follow how severe the inflammation is, not what caused it. What did help to separate the two diseases were clusters of plasma cells and hyaline globules in Kupffer cells.

Part five. The new pathology criteria.

In 2022, the International Autoimmune Hepatitis Pathology Group, seventeen expert liver pathologists working through a structured consensus method called a modified Delphi process, redefined the histology of autoimmune hepatitis. The European guideline of 2025 adopts their criteria.

They describe two patterns that make autoimmune hepatitis likely. The first is the portal pattern. Here there is a mainly portal hepatitis made of lymphocytes and plasma cells, with more than mild interface activity, lobular activity, or both, and no features that suggest another disease. The second is the lobular pattern. Here the inflammation is mainly inside the lobule, with or without necrosis around the central vein, and it is accompanied by at least one of three things: portal hepatitis, interface activity or portal fibrosis. The lobular pattern captures the acute presentation of autoimmune hepatitis.

Emperipolesis and rosettes have been demoted. The consensus says they are not specific for autoimmune hepatitis. They mark severity, not cause. And liver biopsy remains the standard for the diagnosis.

Why does this matter in paediatrics? Because children often present with the acute form of the disease: a sudden hepatitis, sometimes with a high INR, and a biopsy showing lobular inflammation and necrosis around the central vein.

Under older definitions, this picture could be called atypical. Under the 2022 criteria, it now scores as autoimmune hepatitis.

Part six. Autoimmune hepatitis or sclerosing cholangitis?

The most important study on this question comes from King's College Hospital in London. In 2001, Gregorio and colleagues published a sixteen-year prospective study. Every child who presented with autoimmune liver disease had both a liver biopsy and direct imaging of the bile ducts at the time of diagnosis.

Of fifty-five children with autoimmune serology, twenty-seven had an abnormal cholangiogram. They had autoimmune sclerosing cholangitis. So in childhood, sclerosing cholangitis is about as common as autoimmune hepatitis.

Now the key finding. Signs of bile duct disease on the biopsy were present in sixty-five percent of the children with sclerosing cholangitis. But they were also present in thirty-one percent of the children with autoimmune hepatitis whose cholangiogram was normal. So about one biopsy in three from autoimmune hepatitis looks biliary, and about one biopsy in three from sclerosing cholangitis does not. And the disease was not static: eight of the seventeen children with sclerosing cholangitis who had a repeat cholangiogram showed progression.

The practical rule is clear. The biopsy cannot decide between these two diseases. Every child with interface hepatitis and autoantibodies needs a cholangiogram, whatever the biopsy shows. In current practice this means a magnetic resonance cholangiogram at diagnosis.

Part seven. Where the bile sits.

We now turn to cholestasis. When the pathologist sees bile on a slide, the most useful question is not how much bile there is, but where it is. Bile can sit in four places, and there is a fifth pattern in which the ducts themselves are missing.

The first place is the canaliculus. The canaliculus is the tiny channel between two liver cells into which the cell secretes bile. When bile plugs sit in the canaliculi of zone three and there is little inflammation, pathologists call this bland canalicular cholestasis. It means the liver cell cannot push bile out. This is a transport failure. Think of the progressive familial intrahepatic cholestasis syndromes, of drugs, of sepsis, of parenteral nutrition, and of benign recurrent intrahepatic cholestasis.

The second place is the ductule. Ductules are the small channels at the edge of the portal tract that connect the canaliculi to the true bile ducts. When the ductules are dilated and filled with thick, dark bile, this is called ductular or cholangiolar cholestasis. It is strongly associated with sepsis, and the old name for it is cholangitis lenta. In 1982, Lefkowitch described this pattern and showed that it

carries a poor prognosis, and that operating on these patients to look for an obstruction was unrewarding. So when you see this pattern, look for the infection. Do not send the child for a laparotomy.

The third place is the bile duct itself. Bile plugs in the interlobular ducts, together with a ductular reaction and swelling of the portal tract, which pathologists call portal oedema, suggest mechanical obstruction. In an infant, this means biliary atresia until proven otherwise. In an older child, it means large-duct obstruction, for example from a stone or a choledochal cyst.

The fourth pattern is chronic cholestasis. Here the pathologist sees the marks left by bile over a long time: cholate stasis in zone one, copper-associated protein, and liver cells near the portal tract that express CK seven. These features tell you how long the cholestasis has been present. They do not tell you the cause. They are seen in any long-standing biliary disease, including sclerosing cholangitis and Alagille syndrome.

The fifth pattern is ductopenia. Here the portal tracts contain arteries but no bile ducts. We will come back to this in Part eleven. The key message for now is to count the ducts before you conclude that they are missing.

Part eight. The biopsy in biliary atresia.

In an infant with conjugated jaundice, the most urgent question is whether the baby has biliary atresia. The North American and European guideline of 2017 gives a clear position in its recommendation nine. Liver biopsy is the most supportive test for biliary atresia, provided it is taken at the right time and read by a pathologist experienced in paediatric liver disease. The strength of this recommendation is one B.

Which features matter most? The best study is by Russo and colleagues, published in 2016, from the North American Childhood Liver Disease Research Network. They reviewed two hundred and twenty-seven needle biopsies from infants with cholestasis and built a statistical model to separate biliary atresia from all other causes.

Five features stood out. Bile plugs in the ducts or ductules had the strongest association, with an adjusted odds ratio of about thirteen and a half. Portal stromal oedema had an odds ratio of about ten. The absence of duct paucity had an odds ratio of about four. The absence or near absence of blood-forming cells in the liver, which is called extramedullary haematopoiesis, had an odds ratio of about three and a half. And the absence or near absence of giant cells had an odds ratio of about two. Together, these five features separated biliary atresia from

other causes with a concordance index of zero point eight nine, which is very good.

A simple way to remember this is that plugs and oedema say obstruction, while paucity, giant cells and blood-forming cells argue against it.

The same study showed that the typical features grow with age. The ductular reaction, the bile plugs, the fibrosis and small ring-like clusters of liver cells called pseudorosettes all increased in older infants. So a biopsy taken very early can be falsely reassuring.

How good is the biopsy overall? In Russo's study, on blinded review, the diagnostic accuracy was about ninety percent. The sensitivity for biliary atresia was about eighty-eight percent, and the specificity about ninety-three percent.

But there are important mimics. In the same study, fourteen of fifteen biopsies from infants with cholestasis caused by parenteral nutrition were read as obstructive. All three infants with alpha one antitrypsin deficiency of the ZZ type were read as obstructive as well, and cystic fibrosis can also look obstructive. Alagille syndrome, as we will see, can mimic atresia too.

So the practical rule is this. Before calling biliary atresia on a biopsy, make sure the baby has been tested for cystic fibrosis, has had an alpha one antitrypsin phenotype, and has had JAG one testing if Alagille syndrome is possible. If the first biopsy is equivocal, either repeat it or go straight to an intraoperative cholangiogram.

Part nine. Giant cells.

Giant cells are liver cells with many nuclei, formed when several cells fuse. Neonatal giant cell hepatitis used to be treated almost as a diagnosis. It is not. It is a response of the infant liver to stress.

In 2010, Torbenson and colleagues reviewed sixty-two infants at two centres whose biopsies showed neonatal giant cell hepatitis, and followed them to a final diagnosis. About forty-nine percent remained idiopathic. Sixteen percent had hypopituitarism. Eight percent had biliary atresia. Six percent had Alagille syndrome, and six percent had a bile acid synthesis defect. Inflammation was mild or absent in ninety-five percent of the biopsies, and extramedullary haematopoiesis was present in seventy-four percent. Giant cells also appear in twenty to fifty percent of biopsies from babies with biliary atresia.

The lesson is that giant cells mean an infant liver under stress, and the most important treatable causes are endocrine and metabolic. Before calling giant cell hepatitis idiopathic, check the cortisol, the free thyroxine and the thyroid-

stimulating hormone for hypopituitarism, send urine bile acids for a synthesis defect, and make sure biliary atresia and Alagille syndrome have been excluded.

Part ten. The gene behind the pattern.

Many infants with cholestasis now have a genetic diagnosis. A gene panel takes weeks, but the biopsy is available within days. Certain patterns, together with special immunostains, can point to the gene before the sequencing result comes back. The first clue is the GGT. Most of the familial intrahepatic cholestasis syndromes have a low or normal GGT, which is unusual in cholestasis and should always make you think of them.

The first is progressive familial intrahepatic cholestasis type one, caused by variants in the gene ATP eight B one. The GGT is low. The biopsy shows bland canalicular cholestasis with little giant cell change. On electron microscopy, the bile has a coarse, granular look, sometimes called Byler bile.

The second is type two, caused by variants in ABCB eleven, the gene for the bile salt export pump, or BSEP. The GGT is low. The biopsy often shows giant cell hepatitis, necrosis of liver cells in the lobule and fibrosis around the central vein. An immunostain for BSEP is very helpful. In a series by Evason and colleagues, published in 2011, the BSEP stain was absent in ten of twelve patients with this disease.

The third is type three, caused by variants in ABCB four, the gene for MDR three, which moves phospholipid into bile. Here the GGT is high, and the biopsy shows a ductular reaction and portal fibrosis. It looks biliary. An MDR three immunostain and measurement of phospholipid in bile can help.

The fourth is TJP two deficiency. TJP two is a tight junction protein. The GGT is low, the biopsy shows canalicular cholestasis and early fibrosis, and an immunostain for claudin one, a partner protein, is lost.

The fifth is deficiency of the bile acid sensor FXR, caused by variants in the gene NR one H four. The GGT is low. Infants present early with cholestasis, a ductular reaction and rapidly progressive fibrosis. The BSEP stain is absent, because FXR is needed to switch BSEP on. A useful clinical clue is a coagulopathy that does not correct with vitamin K.

The lesson here is that a bland, low GGT cholestasis in an infant is a strong pointer towards a genetic transport defect. Ask the pathologist for the stains, and send the gene panel at the same time.

Part eleven. Counting ducts.

Ductopenia means a loss of the interlobular bile ducts. Paucity of the bile ducts is the term used in infants. It is one of the most important findings in paediatric hepatology, and one of the easiest to get wrong.

The measure is a ratio: the number of interlobular bile ducts divided by the number of portal tracts. In a normal liver, the ratio is between about zero point nine and one point eight. Most portal tracts contain at least one duct.

Many people remember the rule that a ratio below zero point five means paucity. This is the threshold for older children and adults. In a review from 2020, Ayoub and Kamath point out that the threshold is different in young infants. In a term infant, a ratio below zero point nine already suggests paucity. In an older child, the cut-off is below zero point five to zero point seven five. Using the adult cut-off in a young baby will miss real disease.

The denominator matters too. The count is only reliable if there are enough portal tracts. The British guideline of 2020, following the Royal College of Pathologists, sets a minimum specimen of ten millimetres in length containing six portal tracts. For judging duct loss, fewer than eleven portal tracts tends to underestimate the disease. It is also essential not to count ductules as ducts. A ductular reaction at the edge of the portal tract can hide the fact that the true duct in the centre is missing.

A cytokeratin stain, CK seven or CK nineteen, makes the ducts much easier to find. The classic sign of ductopenia is a portal tract that contains a hepatic artery branch with no duct beside it.

The causes of ductopenia in children include Alagille syndrome, alpha one antitrypsin deficiency, Down syndrome, congenital infections such as cytomegalovirus and rubella, hypopituitarism, late biliary atresia, graft-versus-host disease and chronic rejection after transplantation.

Part twelve. Alagille syndrome and the biopsy.

Alagille syndrome is the classic cause of bile duct paucity. It is caused mainly by variants in the gene JAG one, and less often in NOTCH two. It affects the liver, the heart, the eyes, the spine, the kidneys and the face. A natural question is whether paucity is always present on the biopsy. The answer is no, and the published data disagree about when it appears.

In 1999, Emerick and colleagues at the Children's Hospital of Philadelphia described ninety-two patients with Alagille syndrome. Paucity was present in sixty percent of those biopsied before six months of age, and in ninety-five

percent of those biopsied after six months. This suggested that paucity develops with age.

In 2011, Subramaniam and colleagues at King's College Hospital described one hundred and seventeen patients. Their numbers went the other way. Paucity was present in seventy-seven percent of those biopsied before sixteen weeks of age, in seventy-one percent between sixteen weeks and one year, and in sixty-seven percent after one year.

So two large referral cohorts point in opposite directions. We do not know which is correct, and it may depend on how the patients were selected and how the ducts were counted. But the practical lesson is the same either way. In both studies, somewhere between a quarter and two-fifths of infants with Alagille syndrome had no paucity when they were first biopsied. A normal duct count in a young infant does not exclude Alagille syndrome. If you suspect it, send the genetic test. Do not wait for the ducts to disappear.

The King's study also showed how easily Alagille syndrome is mistaken for biliary atresia. Twenty-five percent of the biopsies lacked the characteristic histology. The nuclear medicine scan, which shows whether bile reaches the gut, falsely suggested biliary atresia in twenty-one of thirty-five children, which is sixty percent. Ultrasound falsely suggested atresia in twenty-eight percent. And about nine percent of the children had had a diagnostic laparotomy before they were referred. Two of them had had a Kasai operation.

The 2017 guideline therefore advises that, in an infant with cholestasis, you look for the other features of Alagille syndrome before any intraoperative cholangiogram. Listen for a heart murmur and consider an echocardiogram, which may show peripheral pulmonary stenosis. Ask for an eye examination looking for posterior embryotoxon. Take a spine X-ray looking for butterfly vertebrae. Check the kidneys. A useful bedside clue is a GGT that is very high out of proportion to the bilirubin, together with a ductular reaction on the biopsy. In such an infant, think of Alagille syndrome until JAG one testing says otherwise.

Part thirteen. What the guidelines say.

How do the guidelines compare on liver histology? There are seven points worth knowing.

First, the biopsy in infant cholestasis. Only the paediatric guideline addresses it. The 2017 guideline calls it the most supportive test for biliary atresia, if it is well timed and read by a paediatric pathologist. The adult documents do not cover this question.

Second, the biopsy for autoimmune hepatitis. Here all the guidelines agree. The European paediatric paper says the biopsy is required and makes histology part of the paediatric score. The American guidance says the diagnosis cannot be made without a biopsy. The European adult guideline says the biopsy remains the standard.

Third, emperipolesis and rosettes. Here the guidelines contradict each other. The paediatric scoring and the American guidance still treat them as typical of autoimmune hepatitis, following the 2008 criteria. The pathology consensus of 2022 and the European guideline of 2025 say they are not specific.

Fourth, the lobular or acute form of autoimmune hepatitis. The European documents add something new. They accept a lobular pattern, with or without necrosis around the central vein, as sufficient. The paediatric paper has no lobular criterion. The American guidance reports how common lobular hepatitis and central necrosis are, but does not build them into a pattern.

Fifth, imaging of the bile ducts. This is a material difference. The European paediatric paper and the British guideline recommend a cholangiogram for every child at diagnosis. The American guidance is selective, and recommends it when there is cholestasis, bile duct injury on the biopsy, or inflammatory bowel disease. For children, follow the European position, because of the King's data we discussed in Part six.

Sixth, the adequacy of the specimen. Only the British guideline sets a standard. It asks pathologists to report the length of the specimen, the number of fragments and the number of portal tracts, and sets a minimum of ten millimetres and six portal tracts.

Seventh, the threshold for ductopenia. None of the guidelines gives a numerical threshold. The paediatric guideline lists duct paucity as a feature of Alagille syndrome, but gives no ratio. For the number, you must rely on the pathology literature.

Part fourteen. How an expert reads the report.

Let us bring this together into four habits.

The first habit is to think of the zone first. Name the zone before you name the disease. Zone three necrosis in a teenager may be autoimmune hepatitis, not only paracetamol. And steatohepatitis around the portal tract is the usual paediatric pattern.

The second habit concerns plasma cells. They support autoimmune hepatitis, but their absence does not exclude it. Rosettes and emperipolesis tell you about

severity. And whatever the biopsy shows, every child with autoimmune liver disease needs imaging of the bile ducts.

The third habit is to ask where the bile is. In the canaliculus, think of transport. In dilated ductules, think of sepsis, and look for it. In the ducts, with portal oedema, think of obstruction.

The fourth habit is to count the ducts properly. Count on a CK seven stain, over enough portal tracts. Remember that below zero point nine is abnormal in a term infant. And remember that a normal count does not mean the child does not have Alagille syndrome.

Finally, talk to the pathologist. Tell them the age of the child, the GGT and the question you want answered. The same picture means different things at six weeks and at sixteen years, and in a child with a low GGT and a child with a high one.

Part fifteen. From pattern to action.

To finish the teaching part of the chapter, here are seven common patterns, and what to do with each one.

If the report describes zone three necrosis, think of paracetamol, low blood flow, congestion and acute autoimmune hepatitis. Ask the pathologist about plasma cells and interface activity. Then check drug levels, get an echocardiogram, and send the IgG and autoantibodies.

If the report describes interface hepatitis, think of autoimmune hepatitis, sclerosing cholangitis, Wilson disease, hepatitis B and drugs. Ask about plasma cell clusters, bile duct features and copper. Then arrange a magnetic resonance cholangiogram for every child, and send the ceruloplasmin and urine copper.

If the report describes bile plugs in the ducts with a ductular reaction, think of biliary atresia, and of its mimics: parenteral nutrition, cystic fibrosis, alpha one antitrypsin deficiency and Alagille syndrome. Ask about portal oedema and the duct count. Then send the tests for cystic fibrosis, the alpha one antitrypsin phenotype and JAG one, and plan an intraoperative cholangiogram.

If the report describes giant cell hepatitis, think of idiopathic disease, hypopituitarism, biliary atresia, Alagille syndrome and bile acid synthesis defects. Ask about extramedullary haematopoiesis, the duct count and the BSEP stain. Then send the cortisol, the free thyroxine, the thyroid-stimulating hormone, the urine bile acids and a gene panel.

If the report describes bland canalicular cholestasis and the GGT is low, think of the familial intrahepatic cholestasis syndromes, of TJP two and FXR deficiency,

and of drugs and sepsis. Ask for the BSEP and claudin one stains, and electron microscopy. Then send a cholestasis gene panel.

If the report describes ductular cholestasis, think of sepsis. Ask the pathologist to confirm that the bile is in the ductules and not in the ducts. Then look for the source of infection. Do not send the child for a laparotomy.

If the report describes ductopenia, think of Alagille syndrome, alpha one antitrypsin deficiency, late biliary atresia, graft-versus-host disease and rejection. Ask for a CK seven count and the number of portal tracts. Then send JAG one and NOTCH two testing, and look for the heart, eye and spine features.

Part sixteen. Four clinical cases.

Case one. A fourteen-year-old girl presents with jaundice and an ALT of one thousand two hundred. Her INR is one point nine. Her mother mentions that she took paracetamol for a headache last week, but only in normal doses. The paracetamol level is undetectable. The biopsy shows necrosis around the central veins with lobular inflammation, a mild portal infiltrate with some plasma cells, and no rosettes. The report says the picture is "not typical of autoimmune hepatitis". What do you think? This is the lobular pattern of acute autoimmune hepatitis described by the 2022 pathology consensus. Zone three necrosis in a teenager is not only paracetamol. Send the IgG and the autoantibodies, and arrange a magnetic resonance cholangiogram. If the serology fits, this is likely autoimmune hepatitis, and the absence of rosettes does not argue against it.

Case two. An eleven-year-old boy has positive antinuclear and smooth muscle antibodies, a raised IgG and interface hepatitis rich in plasma cells. The biopsy shows no bile duct injury. The team plans to start steroids and asks whether a cholangiogram is needed, since the biopsy looks like classic autoimmune hepatitis. What do you advise? Yes, he needs a cholangiogram. In the King's study, about half of children with autoimmune serology had sclerosing cholangitis, and one biopsy in three from these children did not look biliary. The biopsy cannot decide between the two diseases. Arrange a magnetic resonance cholangiogram before or soon after starting treatment, because the diagnosis changes the follow-up and the long-term outlook.

Case three. A six-week-old boy has conjugated jaundice and pale stools. The ultrasound does not show a gallbladder clearly. The biopsy shows bile plugs, a ductular reaction and portal oedema, and the pathologist suggests biliary atresia. The duct count is normal. The GGT is over one thousand, far higher than you would expect for his bilirubin. On examination you hear a heart murmur. What should you do before the operation? Consider Alagille syndrome. A normal duct count at six weeks does not exclude it, and Alagille syndrome can mimic atresia

on the biopsy, the nuclear scan and the ultrasound. Ask for an echocardiogram, an eye examination for posterior embryotoxon and a spine X-ray for butterfly vertebrae, and send JAG one testing urgently. Also make sure cystic fibrosis and alpha one antitrypsin deficiency have been excluded. If the features of Alagille syndrome are present, an unnecessary laparotomy can be avoided.

Case four. A two-month-old girl has cholestasis with a normal GGT, poor weight gain and itching. Her biopsy shows giant cell hepatitis with some lobular necrosis. There is extramedullary haematopoiesis and the ducts are present. The report says "neonatal giant cell hepatitis". What next? Giant cells are a response, not a diagnosis. First check the cortisol, the free thyroxine and the thyroid-stimulating hormone to exclude hypopituitarism, and send urine bile acids to look for a synthesis defect. The combination of a normal GGT, giant cells and lobular necrosis should make you think of BSEP deficiency, the second type of familial intrahepatic cholestasis. Ask the pathologist for a BSEP immunostain, and send a cholestasis gene panel. If BSEP is absent, the child needs close follow-up, because this disease can progress quickly and carries a risk of liver cancer in childhood.

Key points to remember.

The acinus explains injury better than the lobule. Blood enters at the portal tract, zone one, and leaves at the central vein, zone three, which has the least oxygen and the most drug metabolism.

Name the zone before the disease. Zone three necrosis means paracetamol, ischaemia, congestion or acute autoimmune hepatitis.

In children, steatohepatitis is often portal, which is called the type two pattern, rather than the adult zone three pattern.

Interface hepatitis is a pattern, not a diagnosis. In children it has five main causes: autoimmune hepatitis, sclerosing cholangitis, Wilson disease, viral hepatitis and drugs.

Plasma cells are present in about two-thirds of biopsies from autoimmune hepatitis. Their absence does not exclude the disease.

Rosettes and emperipolesis follow the severity of inflammation, not its cause. The 2022 pathology consensus no longer treats them as specific.

The 2022 criteria accept a portal pattern and a lobular pattern. The lobular pattern captures the acute presentation common in children.

In children, sclerosing cholangitis is as common as autoimmune hepatitis, and the biopsy cannot tell them apart. Every child needs a cholangiogram.

Where the bile sits matters. Canalicular bile means a transport failure. Ductular bile means sepsis. Bile in the ducts with portal oedema means obstruction.

The biopsy is about ninety percent accurate for biliary atresia. Bile plugs and portal oedema are the strongest features. Parenteral nutrition, cystic fibrosis, alpha one antitrypsin deficiency and Alagille syndrome can all mimic it.

Giant cells are a response of the infant liver. Check for hypopituitarism and bile acid synthesis defects before calling them idiopathic.

In low GGT cholestasis, immunostains for BSEP, MDR three and claudin one can point to the gene before the sequencing result arrives.

Count ducts on a CK seven stain over enough portal tracts. In a term infant, a ratio below zero point nine is paucity. A normal count in a young infant does not exclude Alagille syndrome.

Tell the pathologist the age, the GGT and the question.

Questions to test yourself.

Question one. Why does paracetamol injure zone three? Answer: zone three cells have the most CYP two E one, which turns paracetamol into the toxic product NAPQI, and the least glutathione, which normally neutralises it.

Question two. A biopsy from a child with fatty liver disease shows fat and inflammation mainly around the portal tracts. Does this argue against steatohepatitis? Answer: no. This is the type two pattern, which Schwimmer found in about half of children. It is the usual paediatric pattern, and it is associated with advanced fibrosis.

Question three. A biopsy shows interface hepatitis without an excess of plasma cells. Does this exclude autoimmune hepatitis? Answer: no. Plasma cells are present in only about two-thirds of cases. The diagnosis rests on the whole picture: IgG, autoantibodies, the histological pattern and the response to treatment.

Question four. Why were rosettes and emperipolesis demoted in the 2022 criteria? Answer: when autoimmune hepatitis was compared with hepatitis C of the same grade and stage, they were equally common in both. They reflect how severe the inflammation is, not what caused it.

Question five. A child with autoimmune hepatitis has no bile duct changes on the biopsy. Does the child still need a cholangiogram? Answer: yes. In the King's study, one biopsy in three from children with sclerosing cholangitis showed no bile duct features. Only imaging of the bile ducts can separate the two diseases.

Question six. A septic infant has thick bile in dilated ductules at the edge of the portal tracts. What does this mean, and what should you not do? Answer: this is ductular cholestasis, or cholangitis lenta, a marker of sepsis with a poor prognosis. Look for the source of infection. Do not send the child for a laparotomy to look for an obstruction.

Question seven. Which two biopsy features are most strongly associated with biliary atresia? Answer: bile plugs in the ducts or ductules, and portal stromal oedema, in the study by Russo and colleagues.

Question eight. What is the ductopenia threshold in a term infant, and why is a normal count at six weeks not reassuring? Answer: a ratio of ducts to portal tracts below zero point nine suggests paucity in a term infant. A quarter to two-fifths of infants with Alagille syndrome have no paucity on their first biopsy, so if the clinical picture fits, send JAG one testing.

End of chapter.